

An Interpretation of Quantum Entanglement via the Global Constraint of Energy Conservation—Preserving Local Causality

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Abstract:

Based on the "Space Elementary Quanta (SEQ)" model, this paper proposes a non-local interpretation of quantum entanglement. This interpretation introduces "Global Constraint of Energy Conservation" as the core mechanism, positing that the essence of an entangled state arises from the necessity to comply with a global constraint, while simultaneously satisfying the rules of local entropy increase and Global Constraint of Energy Conservation. This global constraint forces the evolutionary paths of different sites to exhibit coordinated "complementary duality" relationships, thereby giving rise to non-local correlations without any superluminal signal transmission. This framework provides a mechanistic picture for understanding quantum entanglement that does not violate local causality.

Keywords: Quantum Entanglement; Global Constraints; Local Causality; SEQ Model; Emergent Spacetime.

Introduction:

The interpretation of quantum entanglement remains a foundational challenge in modern physics. While numerous theoretical frameworks have been proposed, many that incorporate genuine non-local interactions face a profound conceptual crisis: they risk undermining the principle of local causality, a cornerstone of both General Relativity and Quantum Field Theory. This paper seeks a path that reconciles the undeniable experimental violations of Bell's inequalities—which definitively rule out local hidden variable theories—with the preservation of local causality. We propose a novel interpretive framework wherein quantum correlations do not stem from spooky action-at-a-distance, but emerge from complementary duality solutions across spatially separated sites. These solutions are selected under a global rule constraint, effectively explaining the phenomenon of entanglement without violating the fundamental tenets of local realism.

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[<https://doi.org/10.5281/zenodo.15278220>]. Readers are encouraged to consult the full text for a more comprehensive and logical understanding of the model presented herein.

19.3 Understanding Quantum Entanglement and Non-Local Correlations within This Model —without violating local causality: The essence of an entangled state, therefore, lies in complementary duality solutions across different sites under the global constraint of energy conservation.

Previous sections have explicitly emphasized that this model focuses on local interactions in space and time, and introducing non-local relationships poses significant challenges to the model's computability and physical intuitiveness. However, since quantum entanglement has been widely experimentally verified, the following presents only a conceptual interpretive framework, without delving into detailed analysis.

As discussed in earlier chapters such as Chapter 3 and Chapter 13, the system's evolution is primarily driven by entropy increase and maximum entropy path selection at the quantum level—governed by local energy conduction rules derived from the second law of thermodynamics. So, what then is the nature of entangled states—manifestly non-local correlations—within this model? Do genuine non-local interactions actually exist?

We present a simplified case where the energy states of two distant quantum sites become correlated, without invoking non-local interactions.

Comparison Table of Evolutionary Paths under Local Energy Transfer Rules (as per Chapter 3) using Multiplicative Entropy ($S = \prod m_i$)

Item	Initial State T_1	Path A (State T_2)	Path B (State T_2)
SEQ States	[5, 7, 1, 7, 3]	[5, 6, 2, 6, 4]	[6, 6, 2, 6, 3]
Entropy S	735	1440	1296
Entropy Increase ΔS	--	705	561

Multiple transfer pairs with the maximum energy difference ($7 \rightarrow 1$) exist. Both Path A and Path B satisfy the energy transfer rules mentioned in Chapter 3, including the rule of prioritizing the maximum energy difference between adjacent sites (the maximum entropy path selection rule).

Based on the table above, we can observe that from the T_1 [7,1,7] structure to the T_2 [6,2,6] structure, sites 2 to 4 maintain symmetry. Furthermore, under both potential Path A and Path B at T_2 , sites 2-4 are [6,2,6]. This symmetry arises because the energy difference between sites 2 & 3 and sites 4 & 3 is greater than the energy difference between site 2 & site 1 and site 4 & site 5.

Under this model, as long as the condition for maintaining this symmetry holds, the energy states of site 1 and site 5 are in a correlated state.

When the symmetry from site 2 to site 4 is broken – for instance, measuring site 1 essentially inputs energy into it; if the energy of site 1 becomes greater than that of

site 2 – then at the next moment, this symmetry is broken, and the complementary dual state between site 1 and site 5 is broken.

The above example and discussion equivalently demonstrate that, under the global energy conservation constraint, the local symmetry of the energy distribution generates an energy distribution symmetric configuration. The sites at both ends of this symmetric configuration form an energy state correlation, which is essentially the global energy conservation between site 1 and site 5. Due to the symmetry of the energy distribution from site 2 to site 4, the global energy conservation is compressed into the energy correlation between site 1 and site 5.

As long as the symmetry of the energy distribution configuration is not broken – even if site 1 is measured and a small amount of energy is input, provided the symmetry of the relevant energy distribution configuration in the next evolutionary moment is not broken – site 1 and site 5 remain correlated. Furthermore, although site 1 and site 5 are not adjacent, this correlation requires no non-local interactions.

The explanation for the entangled state provided above should be self-consistent and offer a clear picture. The global constraint of energy conservation fundamentally originates from the postulate of global topological invariance in this model.

This explanation accounts for the correlation of entangled states from the perspective of global energy conservation, without introducing non-local interactions that could potentially invalidate existing theories.

The next step is to see if relevant experimental phenomena can be found to verify or falsify it.

Reference;

1.Zou, Z. K. (2025). Multiplicative Entropy Encodes Time-Causality; Gauge Mediates Mass-Gravity; Frequency Mirrors Metric-Geometry:A Two-Layer Fiber Bundle Chiral Space Model with Topology-Preserving Configuration. Zenodo.
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